Gregory R. Mitchell

Fullstack Software Engineer, Game Developer, Sunglasses Enthusiast

EDITOR TIME, TRAILING 12 MONTHS

1,525 measured hours · 4 hrs 31 mins daily average

TypeScript	33.1%
Vue	16.7%
Markdown	13.7%
PHP	7.9%
Kotlin	6%

ALL-TIME MIX

4,777 measured hours since February 2022 · 3 hrs 3 mins daily average

Java	23.6% · 1,253 h
Kotlin	15.4% · 820 h
TypeScript	13.4% · 713 h
Markdown	8.8% · 469 h
Vue	5.7% · 301 h
PHP	3.3% · 174 h
C	2.7% · 146 h

ALSO SHIPS IN

C++ · JavaScript · Python · Rust

TOOLS

IntelliJ IDEA · WebStorm · VS Code · Git · Gradle · npm · Bun · Docker · Cloudflare · Nuxt

b. 2008

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> **At a Glance**
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> 17,275 commits since 2019 across 119 repositories, and published work going back to 2016, 8,045 of them in 2026 alone. An 88-star top library. 10+ languages shipped in production. Published to 6 package registries: Maven Central, Gradle Plugin Portal, npm, PyPI, Homebrew, and Packagist. 29 patches upstream into 19 organizations I do not own. One published paper.

Software Development

- High Quality Software Development with over 7 years of industry experience, since 2019
- Over 4,700 measured hours of programming, tracked by WakaTime since February 2022:
  - 1,250+ hours in Java
  - 820+ hours in Kotlin
  - 710+ hours in TypeScript
  - 300+ hours in Vue
  - 170+ hours in PHP
  - 140+ hours in C
- Experience in dozens of languages, featuring Java, Kotlin, Vue, C/C++, and TypeScript
- Fan of JetBrains IDEs (such as IntelliJ IDEA and WebStorm), along with Visual Studio Code, where I make JVM, JS, and other framework-related projects

Education

- Dartmouth College | Class of 2030
  - Computer Science major, Quantitative Social Science minor
  - Psychology kept for its research methods
  - Aimed at the mental health and loneliness crisis
  - Matriculating Fall 2026
- High School Graduate | 2026

Experience

- **Computer Science Teacher** | Chicago Pre-College Science and Engineering Program | 2025-2026
  - Wrote and delivered my own curriculum for roughly thirty rising 7th-9th graders, three blocks by grade level, Monday through Thursday
  - Taught Java in 2025 and pivoted through web design, CodeCombat, and an Intro to AI unit in 2026 as each approach met a real constraint
  - Negotiated ~40 donated CodeCombat licenses after the host university's credential policy blocked student logins
  - Imaged and provisioned the classroom Raspberry Pi fleet by hand so lessons were boot-and-execute

Awards

- **Congressional App Challenge** | 1st Place, Illinois 2nd District | 2025
  - Awarded for The Earth App, a platform built to fight the loneliness and mental-health epidemic

Business

- brightplum | 2019-Present
  - Web Development with HTML, CSS and SCSS

    - Uses Frameworks like Jekyll and Gulp
- Team Inceptus | 2020-2024 | Archived
    - Minecraft: Java Edition development platform
    - Heavily utilizes Java, Kotlin, and Spigot API with Netty Channels and NMS
- Calculus Games | 2024-2026 | Archived
    - Game Development Studio
    - Utilizes multiple game engines with Java, Kotlin, C, and C++ for game development
    - Creation of the LevelZ File Format, which is also used often in games
- earth-app | 2025-Present
    - Novelty & Intellectual Curiosity Platform, live in the app stores at v1.0.3
    - Four surfaces, each doing one job: a Drupal 11 API on a VPS for durable state, a Cloudflare Worker for edge/AI/admin, a Nuxt web app, and a native iOS and Android app
    - Utilizes Kotlin Multiplatform as the shared typed model for both the TypeScript frontends and the backend
    - 732 test files across the services that carry a suite
- drupflare | 2026-Present
    - Fourteen repositories answering one question: can Drupal 11 run on Cloudflare Workers?
    - PHP 8.5 statically linked to WebAssembly, executing inside a Durable Object with that object's own SQLite as the database
    - Most ship as standalone libraries, including a Drupal database driver and a documented SQLite layer for Durable Objects
    - v1.0.3 ran 27 pinned production Drupal codebases through a 13-row capability lane and found 104 defects; v1.0.4 (November 2026) verified 94 of 95 module rows and fixed three request-state leaks found by a state audit

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Skills

- **Leadership & Tutoring**
  - 7+ years of experience leading teams of developers, including Team Inceptus and Calculus Games
  - Experience with multiple project management tools, including Jira, Trello, and Asana
  - Exceptional tutoring and mentoring experience with students and peers
  - Effective communication skills, including the ability to explain complex concepts to non-technical audiences
- **Programming**
  - 9+ years of experience in dozens of languages, including Java, Kotlin, C/C++, JavaScript, and more
  - Experience with multiple frameworks, including Spring, Ktor, and Vue
  - Experience with multiple libraries, including Kandy, Lets-Plot, Clikt, and more
  - Designed a custom game engine for the terminal using C and OS-specific libraries
- **Project Management**
  - 8+ years of experience utilizing Git and GitHub for version control and project management
  - Mastered in multiple project management tools, including Jira, Trello, and Asana

Open Source Contributions

Patches merged into repositories I do not own. **29** pull requests across **19** organizations that are not mine, plus **91** issues filed into roughly **45** upstream projects, the first merged when I was 12.

- **raysan5/raylib** | 2025
  - Fixed SHA-1 computation for messages longer than 31 bytes, in one of the most widely used C game libraries

- **actions/setup-java** | 2024
  - Added support for the JetBrains Runtime to GitHub's own official action
- **doxygen/doxygen** | 2025
  - Added Linux ARM to CI
- **SchemaStore/schemastore** | 2023
  - Added the `paper-plugin.yml` schema
- **speechanddebate/tabroom** | 2025
  - Linked tournament names to their information page
- **mfnalex/Spigot-UpdateChecker** | 2023
  - Added HangarMC support
- **js-org/js.org** | 2021
  - Two domain registrations

Maintaining in the other direction: **69** issues opened on my projects by other people and answered, and **19 of 22** outside contributors' pull requests reviewed and merged. MobChip took 13 of those at a 100% merge rate; when I stopped maintaining it, its most active outside contributor forked it as MobChipLite and still ships it today.

Community Roles

- **National Speech & Debate Association** | 2024-Present
  - Proposed new features implemented on the NSDA's tabroom.com platform
  - Created a cross-platform API for tabroom.com using Kotlin Multiplatform and web scraping tools
  - Proved useful during High School Debate Seasons
- **CodeMC** | 2024-Present
  - Designed and developed the Backend API used to automate the developer process, written in Kotlin

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Publications

Retargeting Managed Runtimes to Serverless WebAssembly: A Case Study in Java on Cloudflare Workers

September 2026 | Sole author | 10.6084/m9.figshare.33941974

- Published on figshare and Zenodo under CC BY 4.0, artifact pinned to bytebox v1.0.2 at a specific commit
- Findings from differential testing against a real JVM: startup tracks executable WebAssembly rather than uploaded bytes (a bundle grown 640-fold with inert padding kept startup flat); the documented CPU limit is not the enforced one; 5 of 14 third-party Java libraries run unmodified
- Semantic agreement is not operational equivalence: a correctly translated regular expression cost 14 ms on the reference JVM and 11,010 ms on the host engine for identical output
- Published and citable, not yet refereed; seeking an arXiv endorsement, with a JOSS resubmission due February 2027